

# Thyroid

Hypothyroidism, hyperthyroidism and Graves disease, TSH and T4 genetics, nodules and goiter, and thyroid cancer

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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# 1. Summary

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**Favourable.** Thyroid autoimmunity is on the low side. Graves disease is in percentile 3.3 in the genomic score (and in the 18.8 in the 112 variants). The loci loads of hypothyroidism and Hashimoto (56.2) and hyperthyroidism and Graves (50.1) are typical. The major risk alleles are missing: **PTPN22 R620W G/G**, without the haplotype **HLA-DR3/DQ2.5** (rs2187668 C/C), **SH2B3 C/C**, VAV3 and FLT3 without risk alleles. Hypothyroidism scores (42.2 and 38.2) and Hashimoto scores (33.8 and 51.8) are typical. The local conversion of T4 to T3 is preserved (**DIO2 Thr92Ala T/T**, without the Ala92 allele), and **CHEK2 I157T** is absent.

**Unfavourable.** The structural side of the gland is the weak point. The nodular goiter is in the 90.7 and 93.2 (genomic) percentiles. Thyroid cancer is above the 79th percentile in all scores: 79.1, 82.5, 95.9 (41% coverage) and 80.7 (genomic). The loci load of thyroid cancer is in 93.7. The risk alleles **NRG1 G/G** and **TERT T/T** (2 copies each) and 1 copy in **FOXE1 (A/G)**, **NKX2-1 (C/T)**, DIRC3 and PCNXL2. There is also a slight 1 risk allele in **CTLA4 CT60**, **TPO** and **TSHR** (rs179247).

**What does it mean?** The genetic thermostat tends to a low TSH (score in percentile 19.1, loci load in the 3.9) with a high free T4 (score in percentile 80.1, loci in the 81.7), as already seen in the §43 and the §86. A TSH in the lower half of the reference range may be normal for Sérgio F and, if the T4 and the T3 are free, it is not hyperthyroidism. Genetics makes autoimmune thyroid disease unlikely and puts nodules and differentiated thyroid cancer approximately in the upper quintile. Differentiated thyroid cancer is, in most cases, slow growing and with excellent prognosis. This confirms the §94 (percentile 79.1 thyroid cancer score and high loci load).

## 2. How to read this report

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- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality  $DR2 \geq 0.8$ . Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 23 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 23 of the chip, 0 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 2 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 15 PGS Catalog scores calculated locally with 5.323.160 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ( $n = 633$ ) calculated with exactly the same variants. Percentage  $\geq 80$ : MAIS PROVÁVEL/ALTO;  $\leq 20$ : MENOS PROVÁVEL/BAIXO; between 20 and 80: TÍPICO.
- **Loci load GWAS:** For each trait, all SNPs with  $p \leq 5 \times 10^{-8}$  in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ( $n = 633$ ) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 2677 loci, 2337 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

### 3. Polygenic scores

In the scores of disease, a higher percentile = more risk. In the two levels scores (TSH and T4 free), the percentile indicates only the expected genetic level, without judgment. The scores marked as already used keep the values of previous reports. The Hashimoto of 55 variants was calculated with kNN inference in §18 and is here recalculated with the imputation; the value of the imputation replaces the previous.

#### Thyroid thermostat

| Trait                             | Score (PGS Catalog)                  | Variants used                 | Percentile | Result | Judgement        |
|-----------------------------------|--------------------------------------|-------------------------------|------------|--------|------------------|
| TSH (thyroid stimulating hormone) | PGS004906 (Mulder 2023, 60 variants) | 57 60 (weighted coverage 95%) | 19,1       | BAIXA  | without judgment |
| Free T4                           | PGS004907 (Mulder 2023, 31 variants) | 29 31 (weighted coverage 90%) | 80,1       | ALTA   | without judgment |

- **TSH (thyroid stimulating hormone):** In §43 (Horronas) and in §86.

#### Hypothyroidism and Hashimoto

| Trait                                        | Score (PGS Catalog)                                   | Variants used                               | Percentile | Result | Judgement |
|----------------------------------------------|-------------------------------------------------------|---------------------------------------------|------------|--------|-----------|
| Hypothyroidism                               | PGS000965 (Tanigawa 2022, 4 535 variants)             | 4.043 4.524 (weighted coverage 92%)         | 42,2       | TIPIC  | typical   |
| Hypothyroidism (genomic)                     | PGS005269 (White 2025, PRS-CS, 1.09 million variants) | 1.063.714 1.085.170 (weighted coverage 98%) | 38,2       | TIPIC  | typical   |
| Lymphocytic thyroiditis (Hashimoto)          | PGS005270 (White 2025, 55 variants)                   | 50 55 (weighted coverage 70%)               | 33,8       | TIPIC  | typical   |
| Lymphocytic thyroiditis (Hashimoto, genomic) | PGS005272 (White 2025, PRS-CS, 1.09 million variants) | 1.063.703 1.085.142 (weighted coverage 97%) | 51,8       | TIPIC  | typical   |

- **Hypothyroidism:** In §43 (Horronas).
- **Lymphocytic thyroiditis (Hashimoto):** In §18 (Inflammation and autoimmunity), there with kNN inference; here with the imputation.

#### Hyperthyroidism and Graves

| Trait                            | Score (PGS Catalog)                                   | Variants used                               | Percentile | Result         | Judgement  |
|----------------------------------|-------------------------------------------------------|---------------------------------------------|------------|----------------|------------|
| Hyperthyroidism (thyrotoxicosis) | PGS001042 (Tanigawa 2022, 226 variants)               | 198 221 (weighted coverage 91%)             | 55,5       | TIPIC          | typical    |
| Graves' disease                  | PGS005264 (White 2025, 112 variants)                  | 81 112 (weighted coverage 33%)              | 18,8       | MENOS PROVABLE | favourable |
| Graves disease (genomic)         | PGS005266 (White 2025, PRS-CS, 1.09 million variants) | 1.063.714 1.085.170 (weighted coverage 97%) | 3,3        | MENOS PROVABLE | favourable |

- **Hyperthyroidism (thyroidosis):** In §43 (Horronas).

#### Nodules and goiter

| Trait                   | Score (PGS Catalog)                                   | Variants used                               | Percentile | Result        | Judgement    |
|-------------------------|-------------------------------------------------------|---------------------------------------------|------------|---------------|--------------|
| Benign goiter           | PGS005261 (White 2025, 110 variants)                  | 99 110 (weighted coverage 51%)              | 90,7       | MAIS PROVABLE | unfavourable |
| Benign goiter (genomic) | PGS005263 (White 2025, PRS-CS, 1.09 million variants) | 1.063.714 1.085.170 (weighted coverage 97%) | 93,2       | MAIS PROVABLE | unfavourable |

#### Thyroid cancer

| Trait          | Score (PGS Catalog)                  | Variants used                  | Percentile | Result        | Judgement    |
|----------------|--------------------------------------|--------------------------------|------------|---------------|--------------|
| Thyroid cancer | PGS000087 (Graff 2021, 12 variants)  | 12 12 (weighted coverage 100%) | 79,1       | TIPIC         | typical      |
| Thyroid cancer | PGS005210 (Gibson 2024, 12 variants) | 12 12 (weighted coverage 100%) | 82,5       | MAIS PROVABLE | unfavourable |

| Trait                       | Score (PGS Catalog)                                   | Variants used                               | Percentile | Result        | Judgement    |
|-----------------------------|-------------------------------------------------------|---------------------------------------------|------------|---------------|--------------|
| Thyroid cancer              | PGS004954 (Pozdeyev 2024, 26 variants)                | 20 26 (weighted coverage 41%)               | 95,9       | MAIS PROVABLE | unfavourable |
| Thyroid carcinoma (genomic) | PGS005260 (White 2025, PRS-CS, 1.09 million variants) | 1.063.714 1.085.170 (weighted coverage 97%) | 80,7       | MAIS PROVABLE | unfavourable |

- **Thyroid cancer:** In §24 (Canker Risk) and in §94 (Radiation).

## 4. Key variants by topic

Variants of pituitary-thyroid thermostat, thyroid autoimmunity and papillary cancer replicated loci. All were read directly on the chip. The loci of TSH only inform the expected level of the hormone.

### Thyroid thermostat (TSH, T4 and conversion to T3)

Genes that fix the equilibrium point of TSH and T4 and the conversion of T4 into T3 in tissues.

| Gene   | SNP        | Variant  | Genotype | Source | Effect allele (copies) | Judgement        | Allele effect                                                                                                                                 | Evidence |
|--------|------------|----------|----------|--------|------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------|
| DIO2   | rs225014   | Thr92Ala | T/T      | chip   | C (0 copy/s)           | favourable       | Thr92Ala: C allele (Ala92) reduces local conversion of T4 into T3 in tissues (brain, muscle)                                                  | Moderate |
| DIO1   | rs2235544  |          | C/A      | chip   | C (1 copy/s)           | without judgment | C allele increases the activity of 1 type deiodase: plus free T3 and minus free T4 and reverse T3                                             | Moderate |
| PDE8B  | rs4704397  |          | G/G      | chip   | A (0 copy/s)           | without judgment | Allele A rises to TSH (~0.1 mUI/L per copy); the highest locus of TSH                                                                         | Strong   |
| CAPZB  | rs10917469 |          | A/A      | chip   | A (2 copy/s)           | without judgment | Locus CAPZB of TSH [GWAS Catalog: Thyroid stimulating hormone levels, allele A, beta 0.110 [0.1-0.12] unit increase, p=3E-105, PMID 39067062] | Moderate |
| VEGFA  | rs9472138  |          | C/T      | chip   | T (1 copy/s)           | without judgment | Locus VEGFA of TSH [GWAS Catalog: Thyroid hormone levels, T allele, beta 0.079 [0.059-0.099] unit decrease, p=7E-16, PMID 23408906]           | Moderate |
| PDE10A | rs753760   |          | C/C      | chip   | C (2 copy/s)           | without judgment | Locus PDE10A of TSH [GWAS Catalog: Thyroid hormone levels, C allele, beta 0.100 [0.08-0.12] unit increase, p=1E-24, PMID 23408906]            | Moderate |
| ITPK1  | rs11624776 |          | A/A      | chip   | A (2 copy/s)           | without judgment | Locus ITPK1 of TSH [GWAS Catalog: Thyroid hormone levels, allele A, beta 0.064 [0.042-0.086] unit decrease, p=2E-9, PMID 23408906]            | Moderate |
| TSHR   | rs1991517  |          | C/C      | chip   | G (0 copy/s)           | without judgment | D727E (G allele): associated with slightly lower TSH; without clear clinical effect                                                           | Weak     |

### Thyroid autoimmunity (Hashimoto and Graves)

Immune locks (PTPN22, CTLA4, SH2B3), HLA and the target genes (TSHR, TPO).

| Gene                | SNP       | Variant | Genotype | Source | Effect allele (copies) | Judgement            | Allele effect                                                                                                                 | Evidence |
|---------------------|-----------|---------|----------|--------|------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------|----------|
| PTPN22              | rs2476601 | R620W   | G/G      | chip   | A (0 copy/s)           | favourable           | R620W: the A allele weakens the T lymphocytes' brake; ~1.5-2 more Severe, Hashimoto, 1 type diabetes and rheumatoid arthritis | Strong   |
| CTLA4               | rs3087243 | CT60    | G/A      | chip   | G (1 copy/s)           | unfavourable (light) | CT60: G allele reduces the soluble form of CTLA-4 (immune crossing); ~1.2-1.5x more Graves and Hashimoto                      | Strong   |
| TSHR                | rs179247  |         | A/G      | chip   | A (1 copy/s)           | unfavourable (light) | Intron 1 of TSHR: allele A reduces thymus receptor expression (less tolerance) and is associated with Graves disease          | Strong   |
| HLA-DR3/DQ2.5 (tag) | rs2187668 |         | C/C      | chip   | T (0 copy/s)           | favourable           | Haplotype marker DR3-DQ2.5, the highest risk HLA for Graves, Hashimoto, diabetes type 1 and celiac                            | Strong   |

| Gene  | SNP        | Variant | Genotype | Source | Effect allele (copies) | Judgement            | Allele effect                                                                                                                                            | Evidence |
|-------|------------|---------|----------|--------|------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| SH2B3 | rs3184504  | R262W   | C/C      | chip   | T (0 copy/s)           | favourable           | R262W: T allele is associated with more autoimmunity (hypothyroidism, celiac, type 1 diabetes)                                                           | Moderate |
| VAV3  | rs4915077  |         | T/T      | chip   | C (0 copy/s)           | favourable           | Locus VAV3 of hypothyroidism and autoimmunity of the thyroid [GWAS Catalog: Hypothyroidism, C allele, OR 1.30, p=8E-10, PMID 22493691]                   | Moderate |
| TPO   | rs11675434 |         | C/T      | chip   | T (1 copy/s)           | unfavourable (light) | Locus TPO (thyroid peroxidase): anti-TPO antibodies [GWAS Catalog: Thyroid peroxidase antibody positivity, allele T, OR 1.21, p=2E-16, PMID 24586183]    | Moderate |
| FLT3  | rs76428106 |         | T/T      | chip   | C (0 copy/s)           | favourable           | Intronic variant of FLT3 linked to hypothyroidism [GWAS Catalog: Hypothyroidism, C allele, beta 0.333 [0.29-0.37] unit increase, p=2E-65, PMID 41238958] | Moderate |

## Nodules and thyroid cancer

Loci of thyroid development (FOXE1, NKX2-1), growth (NRG1) and telomerase.

| Gene                | SNP        | Variant | Genotype | Source | Effect allele (copies) | Judgement            | Allele effect                                                                                                    | Evidence |
|---------------------|------------|---------|----------|--------|------------------------|----------------------|------------------------------------------------------------------------------------------------------------------|----------|
| FOXE1 (9q22)        | rs965513   |         | A/G      | chip   | A (1 copy/s)           | unfavourable (light) | Allele A reduces FOXE1 (thyroid development): ~1.7× more copy papillary cancer and also more hypothyroidism      | Strong   |
| NKX2-1/MBIP (14q13) | rs944289   |         | C/T      | chip   | T (1 copy/s)           | unfavourable (light) | The T allele reduces PTCSC3: ~1.3× more papillary cancer per copy                                                | Strong   |
| NRG1 (8p12)         | rs2439302  |         | G/G      | chip   | G (2 copy/s)           | unfavourable         | The G allele is associated with ~1.4× more papillary cancer per copy                                             | Strong   |
| DIRC3 (2q35)        | rs966423   |         | C/T      | chip   | C (1 copy/s)           | unfavourable (light) | DIRC3 locus of thyroid cancer (and TSH) [GWAS Catalog: Thyroid cancer, C allele, OR 1.34, p=1E-9, PMID 22267200] | Moderate |
| PCNXL2 (1q42)       | rs12129938 |         | A/G      | chip   | A (1 copy/s)           | unfavourable (light) | PCNXL2 locus of thyroid cancer [GWAS Catalog: Thyroid cancer, allele A, OR 1.32, p=4E-11, PMID 28195142]         | Moderate |
| TERT                | rs10069690 |         | T/T      | chip   | T (2 copy/s)           | unfavourable         | Intron 4 of TERT: T allele is associated with longer telomeres and more thyroid and glioma cancer                | Moderate |
| CHEK2               | rs17879961 | I157T   | A/A      | chip   | G (0 copy/s)           | favourable           | I157T: CHEK2 reduced function; increased risk of thyroid cancer (~2×)                                            | Strong   |

## 5. GWAS loci burden

Sum of the loci of GWAS Catalog ( $p \leq 5 \times 10^{-8}$ ) of each set. The alleles are oriented towards more risk, or higher level in the sets of TSH and T4, and the sum is compared with the reference population.

| Trait (GWAS Catalog)         | Loci (with genotype) | Sérgio F effect alleles | Calibrated variants | Percentile | Judgement        |
|------------------------------|----------------------|-------------------------|---------------------|------------|------------------|
| TSH (levels)                 | 639 (586)            | 560 de 1172             | 583                 | 3,9        | without judgment |
| Free T4 (levels)             | 85 (78)              | 80 de 156               | 77                  | 81,7       | without judgment |
| Hypothyroidism and Hashimoto | 1262 (1122)          | 1074 de 2244            | 1119                | 56,2       | typical          |
| Hyperthyroidism and Graves   | 261 (196)            | 151 de 392              | 196                 | 50,1       | typical          |
| Goiter and nodules           | 264 (218)            | 200 de 436              | 218                 | 89,7       | unfavourable     |
| Thyroid cancer               | 166 (137)            | 136 de 274              | 137                 | 93,7       | unfavourable     |

## 6. Favourable

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- **Graves' disease:** genomic score in percentile 3.3 and 112 variant score in the 18.8 (33% coverage). **Moderate.**
- **Hypothyroidism and Hashimoto:** typical loci load (56.2) and typical scores (42.2, 38.2, 33.8, 51.8). **Moderate.**
- **PTPN22 R620W G/G**, HLA-DR3/DQ2.5 absent (rs2187668 C/C), **SH2B3 R262W C/C**, **VAV3 T/T** and **FLT3 T/T**: none of the thyroid autoimmunity risk alleles. Strong (PTPN22 and HLA).
- **DIO2 Thr92Ala T/T**: conversion of T4 into normal T3 in the brain and muscle. **Moderate.**
- **CHEK2 I157T absent** (A/A).

## 7. Unfavourable

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- **Thyroid cancer:** percentiles 79.1, 82.5, 95.9 (41% coverage) and 80.7 (genomic); loci load in 93.7. **Moderate.**
- **Benign goiter:** percentiles 90.7 and 93.2 (genomic); loci load of goiter and nodules in 89.7. **Moderate.**
- **NRG1 rs2439302 G/G** and **TERT rs10069690 T/T** (2 copies each); **FOXE1 rs965513 A/G**, **NKX2-1 rs944289 C/T**, DIRC3 and PCNXL2 with 1 copy. Strong for FOXE1, NKX2-1 and NRG1 (each copy ~1.3-1.7×).
- **CTLA4 CT60 G/A**, **TPO C/T** and **TSHR rs179247 A/G**: 1 allele of slight risk of autoimmunity each, compensated by the rest of the profile. **Weak.**
- **Thermostat with low TSH and high free T4:** in mendelian randomization studies, a genetically lower TSH is associated with more goiter, nodules and thyroid cancer, and a higher free T4 is associated with more atrial fibrillation. **Weak to moderate.**

## 8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

| Mechanism                      | Genetic finding                                                                                                        | Meaning             |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------|
| Thyroid alterations            | Thermostat with low TSH (percentile 19.1) and high free T4 (percentile 80.1); nodular goiter above the 90th percentile | Unfavourable (mild) |
| Endocrine dysregulation        | Stable pituitary–thyroid axis and low thyroid autoimmunity; Normal DIO2                                                | Favourable          |
| Somatic Mutation Accumulation  | Papillary cancer appears by somatic mutations (BRAF, RAS) on a high polygenic background (FOXE1, NKX2-1, NRG1)         | Unfavourable        |
| Hayflick's Telomeres and Limit | TERT rs10069690 T/T: longer telomeres, at the expense of increased risk of thyroid cancer and glioma                   | Conditional         |
| Chronic Inflammation           | Without variants leading to autoimmune thyroiditis (PTPN22, HLA-DR3)                                                   | Favourable          |
| Thermogenesis decline          | Local T3 (DIO2) supports the thermogenesis of brown adipose tissue and muscle; preserved conversion                    | Favourable          |
| Genomic instability            | CHEK2 I157T absent; the risk of thyroid cancer is polygenic                                                            | Neutral             |

## 9. Practical suggestions

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All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Thyroid panel in the next routine analysis:** TSH, free T4 and free T3, and anti-TPO antibodies as basal value. Interpreting with genetic thermostat: a low-normal TSH with normal free T4 and T3 is expected. If taking biotin, suspend 2–3 days before harvest, because it interferes with these dosages. *Suggestion (not applied).*
2. **Neck palpation in the annual consultation.** Screening ultrasound in people without symptoms is not recommended in current standards, because it discovers many microcarcinomas that would never cause problems. With goiter and thyroid cancer above the 80–the 95th percentile, a single basal ultrasound can be discussed with the physician; if normal, only signs are repeated. *Suggestion (not applied).*
3. **Signs that justify ultrasound at any time:** palpable nodule, increased volume of the neck, persistent hoarseness, difficulty in swallowing, increased cervical ganglion, or suppressed TSH with high free T4 (which, in this profile, suggests more an autonomous node than Graves). *Suggestion (not applied).*
4. **Adequate iodine, without excess:** about 150 µg/day from fish, shellfish, dairy or iodinated salt. The lack of iodine is the main environmental cause of goiter. Avoid algae or kelp supplements with high doses (> 500 µg/day), which can trigger dysfunction in a nodular thyroid. *Suggestion (not applied).*
5. **Food selenium** (fish, eggs, 1–2 chestnuts occasionally). There is no demonstrated benefit of selenium supplements outside the autoimmune disease, and excess is toxic. *Suggestion (not applied).*
6. **Saving the thyroid to ionizing radiation:** TC of the neck and thorax only when indicated, and thyroid protector on dental radiographs (see §94). Thyroid radiation is the strongest environmental risk factor of papillary cancer. *Suggestion (not applied).*
7. **Information:** with genetically high free T4, any treatment with thyroid hormone should be avoided without clear indication, because the excess of T4 increases the risk of atrial fibrillation and bone loss. *Suggestion (not applied).*

## 10. Limitations

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- The goiter and thyroid cancer scores are based on health records. The lowest coverage (41% and 51%) should be read with caution; the genomic scores, with a coverage of 97%, point in the same direction.
- The chip does not test the rare variants of high effect of medullary thyroid cancer (RET) or family syndromes (PTEN, APC, DICER1). Only family history and a clinical panel exclude them.
- The loci of TSH and T4 explain only a part of the hormonal level; blood analysis always has priority over genetics.
- The load of loci GWAS sums SNPs in linkage imbalance; it is descriptive.
- Conclusions only from genes; it is not a diagnosis.

## 11. Methods and sources

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- **Genotypes:** Sérgio F chip (GRCh38, strand +). The absent SNPs come from local imputation (Beagle 5.5, public panel of complete genomes to 30x), used only with  $DR2 \geq 0.8$ . No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the Imposed VCF; in the palindromics the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed. The alleles of the TSHR loci were manually verified in the literature.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog (15 scores: 2 of hormonal levels, 4 of hypothyroidism and Hashimoto, 3 of hyperthyroidism and Graves, 2 of goiter and 4 of thyroid cancer). Score =  $\Delta$  coefficient  $\times$  dosage. The percentile is empirical in the reference population ( $n = 633$ ), with the same variants.
- **Loci load:** GWAS Catalog ( $p \leq 510^{-8}$ ), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** This study aimed to evaluate the effects of 2023 (TSH and T4), Teumer 2018 (thyroid function), Gudmundsson 2009 and 2012 (FOXE1, NKX2-1, NRG1), White 2025 and Gibson 2024 (scores), Brand 2009 (TSHR), Yuan 2020 (mendelian randomization TSH-cancer), GWAS Catalog and ClinVar.

## 12. Annex: all variants analysed

| Topic                                             | Gene                 | SNP        | Genotype | Source | Effect allele | Copies | Chain                             | Interpretation                                                                |
|---------------------------------------------------|----------------------|------------|----------|--------|---------------|--------|-----------------------------------|-------------------------------------------------------------------------------|
| Thyroid thermostat (TSH, T4 and conversion to T3) | DIO2                 | rs225014   | T/T      | chip   | C             | 0      | strand +                          | without the risk allele                                                       |
| Thyroid thermostat (TSH, T4 and conversion to T3) | DIO1                 | rs2235544  | C/A      | chip   | C             | 1      | strand +                          | 1 copy of effect allele                                                       |
| Thyroid thermostat (TSH, T4 and conversion to T3) | PDE8B                | rs4704397  | G/G      | chip   | A             | 0      | strand +                          | 0 copy of the effect allele                                                   |
| Thyroid thermostat (TSH, T4 and conversion to T3) | CAPZB                | rs10917469 | A/A      | chip   | A             | 2      | strand +                          | 2 copy of the effect allele                                                   |
| Thyroid thermostat (TSH, T4 and conversion to T3) | VEGFA                | rs9472138  | C/T      | chip   | T             | 1      | strand +                          | 1 copy of effect allele                                                       |
| Thyroid thermostat (TSH, T4 and conversion to T3) | PDE10A               | rs753760   | C/C      | chip   | C             | 2      | strand + (palindromic; assumed +) | 2 copy of the effect allele                                                   |
| Thyroid thermostat (TSH, T4 and conversion to T3) | ITPK1                | rs11624776 | A/A      | chip   | A             | 2      | strand +                          | 2 copy of the effect allele                                                   |
| Thyroid thermostat (TSH, T4 and conversion to T3) | TSHR                 | rs1991517  | C/C      | chip   | G             | 0      | without data                      | C/C: without the D727E variant; the TSH receptor is the usual                 |
| Thyroid autoimmunity (Hashimoto and Graves)       | PTPN22               | rs2476601  | G/G      | chip   | A             | 0      | strand +                          | without the risk allele                                                       |
| Thyroid autoimmunity (Hashimoto and Graves)       | CTLA4                | rs3087243  | G/A      | chip   | G             | 1      | strand +                          | 1 copy of risk allele                                                         |
| Thyroid autoimmunity (Hashimoto and Graves)       | TSHR                 | rs179247   | A/G      | chip   | A             | 1      | without data                      | A/G: 1 copy of TSHR intron A associated with Graves disease (moderate effect) |
| Thyroid autoimmunity (Hashimoto and Graves)       | HLA-DR3/ DQ2.5 (tag) | rs2187668  | C/C      | chip   | T             | 0      | strand +                          | without the risk allele                                                       |
| Thyroid autoimmunity (Hashimoto and Graves)       | SH2B3                | rs3184504  | C/C      | chip   | T             | 0      | strand +                          | without the risk allele                                                       |
| Thyroid autoimmunity (Hashimoto and Graves)       | VAV3                 | rs4915077  | T/T      | chip   | C             | 0      | strand +                          | without the risk allele                                                       |
| Thyroid autoimmunity (Hashimoto and Graves)       | TPO                  | rs11675434 | C/T      | chip   | T             | 1      | strand +                          | 1 copy of risk allele                                                         |
| Thyroid autoimmunity (Hashimoto and Graves)       | FLT3                 | rs76428106 | T/T      | chip   | C             | 0      | strand +                          | without the risk allele                                                       |
| Nodules and thyroid cancer                        | FOXE1 (9q22)         | rs965513   | A/G      | chip   | A             | 1      | strand +                          | 1 copy of risk allele                                                         |

| Topic                      | Gene                | SNP        | Genotype | Source | Effect allele | Copies | Chain                             | Interpretation          |
|----------------------------|---------------------|------------|----------|--------|---------------|--------|-----------------------------------|-------------------------|
| Nodules and thyroid cancer | NKX2-1/MBIP (14q13) | rs944289   | C/T      | chip   | T             | 1      | strand +                          | 1 copy of risk allele   |
| Nodules and thyroid cancer | NRG1 (8p12)         | rs2439302  | G/G      | chip   | G             | 2      | strand + (palindromic; assumed +) | 2 risk allele copies    |
| Nodules and thyroid cancer | DIRC3 (2q35)        | rs966423   | C/T      | chip   | C             | 1      | strand +                          | 1 copy of risk allele   |
| Nodules and thyroid cancer | PCNXL2 (1q42)       | rs12129938 | A/G      | chip   | A             | 1      | strand +                          | 1 copy of risk allele   |
| Nodules and thyroid cancer | TERT                | rs10069690 | T/T      | chip   | T             | 2      | strand +                          | 2 risk allele copies    |
| Nodules and thyroid cancer | CHEK2               | rs17879961 | A/A      | chip   | G             | 0      | strand +                          | without the risk allele |